not. This is what we have done with TSP by first developing the methods performing a single action, measured their effectiveness, and then combined them in a single random ensemble.

To make things more concrete, we'll focus on finding an embedding close to the *rectilinear crossing number* (**rcn**) for complete graph K_8: if you remember Guy's conjecture, which we have provided in chapter 15, it gives us an exact formula for this graph's *crossing number* which is 18; in that same chapter, however, we also learned that the rectilinear crossing number for complete graphs can and usually is larger than its crossing number[13]: in this case, `rcn(K8)` is `19`.

So, let's start with a simple step that takes a vertex and slightly moves it within a certain range; after fiddling with the range of update for vertices, we chose to update both `x` and `y` separately, by choosing two random deltas within 10% of the drawing area's width and height respectively; smaller ranges made the algorithm too slow, while with larger values it resulted hectic. Then, running simulated annealing with `k=0.1`, `α=0.97`, `T0=200` and `maxSteps=500`, the average `rcn` over 100 runs was `21.904`. Not bad, but neither particularly good.

Two considerations must be made: first, we kept the number of steps low (if you recall, we used 10000 for TSP in the last section); the second consideration is that, for the same reasons (which we'll discuss in a second), we had to go down from 1000 runs to just 100.